

Designer: hydra-eng | Date: June 2026 | Rev: v2.0 | Based on: polesskiy-dev/iot-risk-data-logger-nfc-samd21 (MIT License)

MCU: U1 – ESP32-C3-MINI-1
3.3V supply | GND | EN pin pulled HIGH via 10k | IO2 strapping to GND
USB-C D+ D- to ESP32-C3 native USB pins
I2C: SDA=GPIO8, SCL=GPIO9 | SPI: CLK=GPIO4, MOSI=GPIO6, MISO=GPIO2

I2C SENSORS (SDA=GPIO8, SCL=GPIO9)
U2: Sensing: SHT40 (Addr 0x70) | Temp, Humidity | VDD=3.3V | 100nF decoupling
U3: Microchip ATECC608A | Addr 0x80 | HW Crypto ECC-P256 | 100nF decoupling
U4: ST ST25DV04K | Addr 0x53 | NFC tag + I2C EEPROM | RF energy harvesting
U9: Maxim MAX17048 | Addr 0x36 | LiPo Fuel Gauge | SDA/SCL shared bus

SPI BUS (CLK=GPIO4, MOSI=GPIO6, MISO=GPIO2)
U5: ADXL345 Accelerometer | CS=GPIO5 | ±16g range, full resolution, 100Hz ODR
U6: SX1262 LoRa Module | CS=GPIO7, BUSY=GPIO3, DIO1=GPIO10, RST=GPIO1
U8: 65kHz eInk panel | J9: 8-pin 1.54mm pitch Bm D+ D- USB-C to eInk panel connector
NOTE: 50-ohm impedance-controlled trace required from module to U.FL
J3: e-ink Display FPC 8-pin | CS=GPIO19, DC=GPIO18, RST=GPIO17, BUSY=GPIO16
B 1.54" 200x200 B/W | SSD1681 driver IC | zero-power image retention

GPS MODULE (UART)
U7: u-blox MAX-M10S | TX=GPIO21 (ESP RX), RX=GPIO20 (ESP TX) | 9600 baud
Power: LDO (U12: AMS1117-3.3) for GPS power domain
LDO enable: GPIO controlled for switchable power-off
Backup battery pad: CR1220 coin cell for RTC retention
Passive antenna or active antenna with LNA bias

POWER SYSTEM
J1: JST-PH 2-pin LiPo connector (1000mAh 3.7V)
J2: USB-C receptacle (TP4056 VBUS + ESP32 D+/D-)
U9: TP4056 LiPo charger
U10: DW01A + FS8205 battery protection IC | overvoltage/overcurrent/reverse
U11: AMS1117-3.3 | LiPo+ -> VBAT -> AMS1117 IN -> 3.3V -> all ICs
D4: LED + R4(330ohm) battery indicator driven from GPIO15

INDICATORS, BUZZER & TAMPER
RGB LED (common cathode): GPIO12 -> R1(330) -> D1(RED) -> GND | GPIO13 -> R2(330) -> D2(GRN) -> GND | GPIO14 -> R3(330) -> D3(BLU) -> GND
BUZZER: GPIO11 -> R6(1k) -> Q1(2N2222 NPN BASE) | Q1_COLLECTOR -> BZ1 -> 3.3V | Q1_EMITTER -> GND
TAMPER: GPIO0 -> R5(10k) -> 3.3V (pullup) | PCB battery: GPIO0 -> lid_pad -> GND via conductive trace
Lid CLOSED: trace intact, GPIO0=LOW | Lid OPEN: trace broken, GPIO0=HIGH -> TAMPER EVENT

Sheet: 1
File: cargopulse_v2.kicad_sch
Title:
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.1 Id: 1/1